SECTIONS OF UNRAMIFIED AND SEPARATED MORPHISMS ARE OPEN RESPECTIVELY CLOSED IMMERSIONS, AND BOTH FOR UNRAMIFIED SEPARATED MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file etale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Source reference. Original reference 01KK: sources/stacks-chow/tags.txt, source lines 2053–2053. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02GE: sources/stacks-chow/tags.txt, source lines 3225–3225. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01JY: sources/stacks-chow/tags.txt, source lines 2031–2031. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Proposition 0.1. *Sections of unramified morphisms.*

(1) *Any section of an unramified morphism is an open immersion.*
(2) *Any section of a separated morphism is a closed immersion.*
(3) *Any section of an unramified separated morphism is open and closed.*

Proof. Fix a base scheme S. If $f : X' \to X$ is any S-morphism, then the graph $\Gamma_f : X' \to X' \times_S X$ is obtained as the base change of the diagonal $\Delta_{X/S} : X \to X \times_S X$ via the projection $X' \times_S X \to X \times_S X$. If $g : X \to S$ is separated (resp. unramified) then the diagonal is a closed immersion (resp. open immersion) by Schemes, Definition 01KK (resp. Morphisms, Lemma 02GE). Hence so is the graph as a base change (by Schemes, Lemma 01JY). In the special case $X' = S$, we obtain (1), resp. (2). Part (3) follows on combining (1) and (2). □

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